HUM1012 : Logic and Language Structure

(LT)

Slot : B21 (Mon 04.25 pm – 05.55 pm) + E14 (Tue 02.50 pm – 04.20 pm)
D21 (Tue 01.15 pm – 02.45 pm) B14 + (Wed 02.50 pm – 04.20 pm)

Venue : AB02 – 301 (B21+E14)
AB02 – 301 (B14+D21)

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MODULE 4 Logic in Natural Language and understanding

Natural Language semantics 6

Propositional logic (truth value, truth table, negation, conjunction, disjunction, conditional, bi-conditional),

formulas (atomic and complex) and propositional letters, equivalence, contradiction, tautology, entailment, implicature, and presupposition in relation to set theory, relation, function, and logic.

COURSE OUTCOMES:

CO4: Understanding the basics of mathematical logic and its application to natural language semantics.

4.1 Propositional logic

(truth value, truth table, negation, conjunction, disjunction, conditional, bi-conditional)

- Propositional Logic (also called Boolean logic or sentential logic).
- It is the branch of logic that deals with propositions (statements) that can be either True (T) or False (F).

(i) Proposition

- A proposition is a declarative statement that is either true or false, but not both.

Example:

"The sky is blue." → True (T) "2 + 2 = 5." → False (F)

Truth value

- The value assigned to a proposition.

Example:

T (True) F (False)

(ii) Truth Tables & Logical Connectives

- Propositions can be combined using connectives.
- Each connective has a truth table.

(a) Negation (¬P / NOT P)

- Negation reverses the truth value.

P	¬P
T	F
F	T

Example:

Proposition (P): "It is raining."

Negation (¬P): "It is NOT raining."

P (It is raining)	¬P (It is NOT raining)
T	F
F	T

(b) Conjunction (P ^ Q / AND)

- True only if both are true.

P	Q	P ∧ Q
T	T	T
T	F	F
F	T	F
F	F	F

Example:

P: "I will go jogging."

Q: "I will carry an umbrella."

P ∧ Q: "I will go jogging AND I will carry an umbrella."

P	Q	P ∧ Q	Meaning
T	T	T	Jogging AND Umbrella
T	F	F	Jogging but NO Umbrella
F	T	F	No Jogging, Umbrella
F	F	F	Neither Jogging nor Umbrella

(c) Disjunction (P ∨ Q / OR)

- True if at least one is true. (Inclusive OR).

P	Q	P ∨ Q
T	T	T
T	F	T
F	T	T
F	F	F

Example:

P: "I will study."

Q: "I will play football."

P ∨ Q: "I will study OR play football (or both)."

P	Q	P ∨ Q	Meaning
T	T	T	Study and Play
T	F	T	Study only
F	T	T	Play only
F	F	F	Neither study nor play

(d) Conditional (P → Q / If P then Q)

- If P is true, Q must also be true. False only when P = T and Q = F.

P	Q	P → Q
T	T	T
T	F	F
F	T	T
F	F	T

Example:

P: "If it rains (P)."

Q: "The ground will be wet (Q)."

P → Q: "If it rains, then the ground will be wet."

P (Rain)	Q (Wet Ground)	P → Q	Meaning
T	T	T	Rain causes wet ground
T	F	F	Rained, but ground not wet (contradiction)
F	T	T	Did not rain, but ground wet (possible)
F	F	T	Did not rain, ground not wet

(e) Bi-Conditional (P ↔ Q / P if and only if Q)

- True if both have the same truth value.

P	Q	P ↔ Q
T	T	T
T	F	F
F	T	F
F	F	T

Example:

P: "I will pass the exam."

Q: "I will study hard."

P ↔ Q: "I will pass if and only if I study hard."

P (Pass Exam)	Q (Study Hard)	P ↔ Q	Meaning
T	T	T	Passed because studied
T	F	F	Passed without studying
F	T	F	Studied but still failed
F	F	T	Did not study and did not pass

Types of Propositions

Atomic Proposition

Simple, indivisible.

Example: "It is raining."

Compound Proposition

Formed by connecting atomic propositions with logical connectives.

Example: "It is raining AND it is cold." ($P \wedge Q$)

Tautology

Always true regardless of truth values.

Example: ($P \vee \neg P$)

Contradiction

Always false.

Example: ($P \wedge \neg P$)

Contingency

Sometimes true, sometimes false depending on P and Q.

Example: ($P \wedge Q$)

Tools for Propositional Logic

- Truth Tables → Manual reasoning
- Boolean Algebra → Simplifying logical expressions
- SAT Solvers (Satisfiability solvers) → Automated checking of satisfiability
- Prolog → Logic programming language
- Symbolic Logic Calculators (like WolframAlpha, Logicly, Tarski's World)
- Automated Theorem Provers (Z3, Coq, Isabelle)

Applications of Propositional Logic

Computer Science

- Digital circuits (logic gates use AND, OR, NOT, etc.)
- Software verification
- Database queries (SQL uses AND, OR, NOT)
- Programming (conditional statements if-else)

Artificial Intelligence

- Knowledge representation
- Reasoning systems
- Expert systems (medical diagnosis, legal reasoning)

Mathematics

- Proof construction
- Formal verification of theorems

Everyday Decision Making

- Reasoning with rules:
  - "If it rains, then I will carry an umbrella."

4.2 Logic, Set Theory, Relation, Function

in relation with

Formulas (atomic and complex) and propositional letters, equivalence, contradiction, tautology, entailment, implicature, and presupposition

Atomic & Complex Formulas

Logic:

Atomic: simplest propositions like P, cannot be broken further.

Complex: built using logical connectives (¬, ∧, ∨, →, ↔).

Set Theory:

Atomic = basic subset of universe U.

Complex = formed using set operations (union, intersection, complement).

Relation:

Atomic = direct relation between proposition and truth.

Complex = combination of relations (like composition or union of relations).

Function:

Atomic = truth assignment function $v(P) \in \{0,1\}$.

Complex = Boolean functions combining inputs (max, min, 1−x).

Propositional Letters

Logic:

- ✓ Propositional letters are symbols like P, Q, R.
- ✓ Each one stands for a whole atomic statement (cannot be broken down further in propositional logic).

  Example:

  P: “It is raining.” Q: “The ground is wet.”

Set Theory:

- ✓ Each propositional letter corresponds to a subset of the universe of possible worlds/cases where that proposition is true.

  Example:

  Universe U = {w1,w2,w3} , P={w1,w3} (true in w1, w3) , Q={w2,w3}}

Relation:

- ✓ A propositional letter can be seen as a binary relation between a possible world and a truth value.

Example:

(w1, 1), (w2, 0), (w3, 1) for P.

Function:

- ✓ A propositional letter is a truth assignment function:

$$v(P) : U \rightarrow \{0,1\}$$

Example:

$v(P)\ (w1) = 1, \quad v(P)\ (w2) = 0, \quad v(P)\ (w3)=1.$

Equivalence

Logic:

Two formulas are equivalent ($\varphi \leftrightarrow \psi$)

if they have the same truth values under all interpretations.

Set Theory:

Equivalent formulas correspond to the same subset of U.

Equality of sets (A = B).

Relation:

Equivalence relation (reflexive, symmetric, transitive).

Function:

Truth-value functions of φ and ψ are identical (bijection of outputs).

Contradiction

Logic:

A formula always false ($P \wedge \neg P$).

Set Theory:

Equivalent to the empty set ($\emptyset$), since no case satisfies it.

Relation:

Empty relation (no pairs).

No pairs exist ($\emptyset \subseteq A \times B$).

Function:

Constant-zero function $f(x) = 0$.

Tautology

Logic:

A formula always true ($P \vee \neg P$).

Set Theory:

Equivalent to the universal set (U).

Relation:

Universal relation (all pairs allowed).

$(A \times B)$

Function:

Constant-one function $f(x) = 1$.

Entailment

Logic:

$\Phi \models \psi$ if every truth assignment that makes φ true also makes ψ true.

(or) $\Gamma \models \varphi$

Set Theory:

Corresponds to subset inclusion ($A \subseteq B$).

Relation:

Partial order relation ($\subseteq$, reflexive + transitive).

Function:

Monotonic functions: preserves truth under extension.

truth of φ guarantees truth of ψ.

Implicature

Logic:

A pragmatic inference (not strict).

Weaker than entailment (e.g., “Some students passed” → implies “Not all passed,” but logically could still be “all”).

Set Theory:

“Probable superset” — not guaranteed equality.

Suggests a “likely” superset or restricted interpretation, not strict equality.

Relation:

Defeasible relation (can be overridden by context).

Function:

Non-deterministic mapping: one input can lead to multiple possible inferences.

Presupposition

Logic:

Background condition needed for a statement

(e.g., “The king of France is bald” presupposes that France has a king).

Set Theory:

Requires non-empty domain ($\neq \emptyset$) of interpretation before formula evaluation.

Relation:

Relation is only defined on a subset of domain (restricted domain).

Function:

Partial function — undefined if presupposition is not satisfied.

Comparison Logic Vs Set Theory Vs Relation Vs Function

Concept	Logic	Set Theory	Relation	Function
Atomic Formula	Basic indivisible proposition 'P'	Subset of universe $A \subseteq U$	Relation between element and membership ($x \in A$)	Truth assignment $v(P) \in \{0,1\}$
Propositional Letters	Atomic formulas (P, Q, R)	Subsets of a universe	Pairs of (world, truth value).	Truth assignment function $v(P) : U \rightarrow \{0,1\}$.
Complex Formula	Built from connectives ($\neg, \wedge, \vee, \rightarrow, \leftrightarrow$) (e.g. $P \vee Q$)	Set operations: union, intersection, complement (e.g. $A \cup B$)	Composition / union of relations	Boolean function f(x, y) (e.g., AND = min, OR = max).
Equivalence	Always same truth values (e.g. $\Phi \leftrightarrow \psi$)	Equality of sets A=B	Equivalence relation (reflexive, symmetric, transitive)	Bijection of truth sets
Contradiction	Always false (e.g. $P \wedge \neg P$)	Empty Set $\emptyset$	Empty relation (no pairs)	Constant 0 function

Contd..

Concept	Logic	Set Theory	Relation	Function
Tautology	Always true (e.g. P ∨ ¬P)	Universal Set U	Universal relation (A × B)	Constant 1 function
Entailment	truth of φ guarantees ψ Φ ⊨ ψ (⊨ semantic entailment)	Subset relation A ⊆ B	Partial order (⊆)	Monotonic function (truth-preserving)
Implicature	Pragmatic inference (weaker than entailment) ("Some ⇒ not all")	Probable/likely superset (not strict)	Defeasible relation Probable/likely superset (not strict)	Non-deterministic mapping (multiple outputs possible)
Presupposition	Background assumption "King of France is bald" -> France have a king	Non-empty domain needed	Relation defined only on restricted Domain	Partial function (undefined if precondition fails)

Summary of the “types”

Atomic vs Complex → Building blocks vs combinations.

Equivalence → Two formulas mean the same (set equality, bijection).

Contradiction → Always false (empty set, zero function).

Tautology → Always true (universal set, one function).

Entailment → Guarantee relation (subset, monotone function).

Implicature → Pragmatic inference (default, defeasible).

Presupposition → Background requirement (domain restriction, partial function).

Logic, Set Theory, Relation, Function Tools and Applications

Concept	Tools	Applications in NLP
Logic	- **Prolog** (symbolic reasoning) - **NLTK logic module** (first-order logic parser) - **Coq, Isabelle** (formal verification) - **AllenNLP** (semantic role labeling with logic grounding)	- **Semantic parsing** (mapping natural language to logical forms) - **Textual entailment** (e.g., "John is a man" ⇒ "John is human") - **Question answering** using logical reasoning - **Natural language inference (NLI)** tasks in GLUE/SNLI datasets
Set Theory	- **Python sets** for representing word senses, meanings - **spaCy, NLTK** (lexical resources like WordNet) - **Pandas/Numpy** for set-based operations	- **Word sense disambiguation** (synonyms as sets) - **Information retrieval** (documents as sets of terms) - **Bag-of-words models** (set membership of words) - **Ontology building** (concepts as sets of entities)

Contd..

Concept	Tools	Applications in NLP
Relations	- **Graph databases** (Neo4j, RDF, OWL) - **Stanford CoreNLP dependency parser** - **NetworkX** for relation graphs - **OpenIE** (extracting relations)	- **Dependency parsing** (subject–verb–object triples) - **Knowledge graphs** (DBpedia, Wikidata) - **Relation extraction** ("Barack Obama → born_in → Hawaii") - **Coreference resolution** (linking mentions to entities)
Functions	- **Lambda calculus tools** (Haskell, Python λ) - **Montague grammar parsers** - **PyTorch/TensorFlow** (functions as neural mappings) - **NLTK semantic parser**	- **Compositional semantics** (sentence meaning = function applied to arguments) - **Quantifiers** ("every", "some") modeled as functions over sets - **Machine translation** (mapping functions from one language to another) - **Text generation** (functions from latent space → words) - **Mappings in NLP** (word → embedding)

4.3 Formulas (atomic and complex)

- Formulas are the building blocks of logical reasoning.
- In logic, a formula is a well-formed expression built from:
  - ✓ Propositional letters / variables (e.g., P, Q, R)
  - ✓ Logical connectives (¬, ∧, ∨, →, ↔)
  - ✓ Quantifiers (∀, ∃) in predicate logic
  - ✓ Parentheses to indicate structure.
  - ✓ Variables, functions, predicates (in first-order logic).

Atomic formulas = indivisible statements.

- Atomic formulas are building blocks (membership, relation, equality).

Complex formulas = combinations of atomic formulas.

- Complex formulas are logical combinations that mirror set operations, relation composition, and function properties.

(i) Atomic Formulas

- An atomic formula (or atom) is the simplest type of formula.
- They cannot be broken down into smaller formulas.
- Represents basic facts.
- Atomic formulas are like basic meaning units extracted from sentences, often corresponding to predicates or relations.

Atomic formulas = basic statements about the world

Examples:

Propositional Logic:

- P, Q, R (just atomic propositions).

Example:

p = "It is raining.“ q = "The ground is wet."

Predicate Logic:

Studies(John, LLS) = John studies LLS.

Exam(LLS) = “LLS exam”

Atomic Formulas in,

Set theory:

An atomic formula like $x \in A$ states element-set membership.

Relation:

R(x, y) means (x, y) belongs to relation 'R'.

Function:

f(x) = y is an atomic statement about function application.

Logic:

Direct truth-value statements like P = True or False.

(ii) Complex Formulas

- Formed by combining atomic formulas with logical connectives or quantifiers.
- They can express relationships and reasoning.

Example:
p = "It is raining." q = "The ground is wet."

Types of Complex Formulas:

Negation (¬ϕ): "Not ϕ"

Example: ¬p = "It is not raining."

Conjunction (ϕ ∧ ψ): "ϕ and ψ"

Example: p ∧ q = "It is raining AND the ground is wet."

Disjunction (ϕ ∨ ψ): "ϕ or ψ"

Example: p ∨ q = "It is raining OR the ground is wet."

Implication/Conditional (ϕ → ψ): "If ϕ then ψ"

Example: p → q = "If it is raining, then the ground is wet."

Equivalence/Biconditional (ϕ ↔ ψ): "ϕ if and only if ψ"

Example: p ↔ q = "It is raining if and only if the ground is wet."

Quantified Formulas (Predicate Logic):

Universal quantification: ∀x Studies(x) = "Everyone studies."

Existential quantification: ∃x Studies(x, LLS) = "Someone studies LLS."

Complex Formulas in,

Set theory:

$$x \in A \wedge x \in B \implies x \in A \cap B$$

$$x \in A \vee x \in B \implies x \in A \cup B$$

Relation:

$R(x,y) \wedge S(y,z)$ $R(x,y)$ → chaining relations (composition).

$\forall x \exists y$ $R(x,y)$ → every element has a related partner.

Function:

$f(x) = y \wedge g(y) = z$ → function composition.

Logic:

Formulas combine atomic statements into arguments with logical structure.

Types of Formulas in Logic & Natural Language

Type	Logic Form	Natural Language Example
Atomic	P(x) (or) Q(x)	"Mary runs."
Negation	¬P(x)	"Mary does **not** run."
Conjunction	P(x) ∧ Q(x)	"Mary runs **and** sings."
Disjunction	P(x) ∨ Q(x)	"Mary runs **or** sings."
Implication	P(x) → Q(x)	"If Mary runs**,** she gets tired."
Equivalence	P(x) ↔ Q(x)	"Mary passes **if** she studies."
Universal Quantification	∀x P(x)	"**Every** student studies."
Existential Quantification	∃x P(x)	"**Some** student studies."

Types of Formulas in Logic, Set Theory, Relation and Function

Formula Type	Definition	Example in Logic	Set Theory Equivalent	Relation Equivalent	Function Equivalent
Atomic	Basic indivisible formula	$P(x)$	$x \in A$	$(a, b) \in R$	$f(a) = b$
Negation	Not of a formula	$\neg P(x)$	$x \notin A$	$(a, b) \notin R$	$f(a) \neq b$
Conjunction	Both must hold	$P(x) \wedge Q(x)$	$x \in A \cap B$	$(a, b) \in R \wedge (a,b) \in S$	$f(x)=y \wedge g(x)=z$
Disjunction	Either must hold	$P(x) \vee Q(x)$	$x \in A \cup B$	$(a,b) \in R \vee (a,b) \in S$	$f(x)=y \vee g(x)=z$
Implication	If...then...	$P(x) \rightarrow Q(x)$	$x \in A \Longrightarrow x \in B$	$R \subseteq S$	If $f(x)=y$ then $g(x)=z$
Equivalence	Both directions	$P(x) \leftrightarrow Q(x)$	$A = B$	$R = S$	$f = g$
Universal	For all	$\forall x\, P(x)$	$A = U$ (whole universe)	Every ‘a’ has relation ‘b’	Function is total
Existential	There exists	$\exists x\, P(x)$	$A \neq \emptyset$	Some ‘a’ related to ‘b’	Function is non-empty

4.4 Propositional Letters

- In Propositional Logic (PL), a propositional letter (also called an atomic proposition or sentence letter).
- They can be atomic or part of complex formulas.
- It is a symbol (like P, Q, R,...) that stands for a basic statement.
- Each propositional letter can be True (T) or False (F).
- They are the building blocks of logical formulas.
- Every propositional letter is basically a placeholder for a property, membership, relation, or function statement.

Example:

P: "It is raining."

Q: "The ground is wet."

R: "John is happy."

Types of Propositional Letters

(A) Based on Structure

Atomic Propositional Letters

- ✓ The simplest units.
- ✓ Cannot be broken down further.
- ✓ Example: P, Q, R.

Complex Propositions (formed using letters + connectives)

- ✓ Built from atomic letters using logical connectives.
- ✓ Example: $(P \wedge Q)$, $(P \rightarrow R)$.
- ✓ These are formulas, but they originate from propositional letters.

(B) Based on Truth Value

Tautological Letters (always true in meaning)

- ✓ If a propositional letter stands for a statement that is universally true.
- ✓ Example: Let P = "2 is even." (always true).

Contradictory Letters (always false in meaning)

- ✓ If it represents something always false.
- ✓ Example: Q = "5 is less than 2." (always false).

Contingent Letters (sometimes true, sometimes false)

Truth depends on circumstances.

Example: R = "It is raining." (can be true or false).

Type of Propositional Letter	Example Symbol	Example Meaning	Truth Nature
Atomic	P	"It is raining"	Single indivisible
Complex (via connectives)	(P∧Q)	"It is raining AND cold"	Built formula
Tautological	P	"2 is even"	Always true
Contradictory	Q	"5 < 2"	Always false
Contingent	R	"John is sleeping"	Sometimes true/false

Propositional letters in,

Logic:

Raw truth symbols used to build formulas.

Propositional letters are truth-valued symbols.

Set theory:

Capture membership.

A propositional letter like P(x) means "$x \in A$".

Relations:

Capture pair belonging.

R(x, y) can be seen as a propositional letter meaning "$(x, y) \in R$".

Functions:

Capture input-output pairing.

F(x, y) as a propositional letter meaning "f(x) = y".

Type	In Logic	In Set Theory	In Relations	In Functions
Atomic / Simple	A basic proposition like 'P'	Statement like $x \in A$	Statement like $(a, b) \in R$	Statement like $f(x) = y$
Compound	Built using connectives $(\wedge, \vee, \neg, \rightarrow)$	$X \in A \cap B$, $x \in A \cup B$	$(a, b) \in R \wedge (a, b) \in S$	$f(x)=y \wedge g(x)=z$
Tautological	Always true $(P \vee \neg P)$	$A \cup A^c = U$ (universe)	$R \cup R^c = U \times U$	Identity mapping always defined
Contradictory	Always false $(P \wedge \neg P)$	$A \cap A^c = \emptyset$	$R \cap R^c = \emptyset$	Impossible function value $f(x)=y \wedge f(x) \neq y$
Contingent	Sometimes true, sometimes false	$x \in A$ depends on set	$(a, b) \in R$ may hold or not	$f(x) = y$ depends on input
Universal	$\forall x\, P(x)$	$A = U$	All pairs exist $(R = U \times U)$	Total function (defined for all inputs)
Existential	$\exists x\, P(x)$	$A \neq \emptyset$	At least one pair exists	Function is non-empty (domain not void)

Example

If we define:

P(x): “$x \in A$”

Q(x): “$x \in B$”

R(x,y): “$(x,y) \in R$”

F(x,y): “$f(x)=y$”.

Then:

$P(x) \wedge Q(x) \rightarrow x \in A \cap B$ (set theory)

$R(x, y) \vee R(y, x) \rightarrow$ symmetric relation check

$\forall x \exists y \; F(x, y) \rightarrow$ definition of a function.

4.5 Equivalence

- Equivalence means two things are the same in some meaningful sense.
- In logic, it means two formulas always have the same truth value.
- In sets, it means two sets have the same members.
- In relations, it means a relation satisfies reflexivity, symmetry, and transitivity.
- In functions, it means two functions behave identically on their domain.

Logic → equivalence = same truth conditions.

Sets → equivalence = same elements or cardinality.

Relations → equivalence = reflexive + symmetric + transitive.

Functions → equivalence = identical outputs or structural sameness.

(i) Equivalence in Logic

- Two formulas P and Q are logically equivalent if $P \leftrightarrow Q$ is a tautology (always true).
- Notation: $P \equiv Q$.

Example:

$\neg(P \vee Q) \equiv \neg P \wedge \neg Q$ (De Morgan's Law).

Types in Logic:

- ✓ Tautological equivalence → True by logic laws.
- ✓ Contradictory equivalence → Always false ($P \equiv \neg P$).
- ✓ Contingent equivalence → True in some cases, false in others.

(ii) Equivalence in Set Theory

Set equivalence:

- Two sets A and B are equivalent if they have the same elements.

$$A = B \iff (\forall x) \quad (x \in A \leftrightarrow x \in B).$$

Cardinality equivalence:

- Two sets A and B are equivalent if there is a bijection $f: A \rightarrow B$.

Types in Sets:

- ✓ Equal sets (A = B)
- ✓ Equivalent cardinality (|A|=|B|)
- ✓ Subset equivalence ($A \subseteq B$ and $B \subseteq A$).

(iii) Equivalence in Relations

- An equivalence relation 'R' on a set 'A' satisfies:
  - ✓ Reflexive: (a, a) ∈ R. $a \equiv a \pmod{n}$
  - ✓ Symmetric: If (a, b) ∈ R, then (b, a) ∈ R. $a \equiv b \pmod{n} \Rightarrow b \equiv a \pmod{n}$
  - ✓ Transitive: If (a, b) ∈ R and (b, c) ∈ R , then (a, c) ∈ R.

    $b \equiv c \pmod{n} \Rightarrow a \equiv c \pmod{n}$

Example:

Two integers are congruent modulo n if their difference is divisible by n, meaning they leave the same remainder when divided by n.

"Congruence modulo n" on integers.

$15 \equiv 7 \pmod{4}$ because 15−7=8, divisible by 4.

Types of equivalence relations:

- ✓ Identity relation (every element only related to itself).
- ✓ Congruence relation (partitioning integers).
- ✓ Isomorphism relation (structural equivalence in algebra/graphs).

(iv) Equivalence in Functions

- Two functions f, g : A→B are equivalent if:

Pointwise equivalence:

f(x) = g(x) for all x ∈ A.

Example: **f(x) = 2x and g(x) = x + x**.

Functional equivalence by composition:

f ∘ h = g ∘ h (or) h ∘ f = h ∘ g for some transformation 'h'.

Isomorphic equivalence:

There exists a bijection 'h' such that **g = h ∘ f ∘ h–1**.

∘ Composition Operator

Comparison Table

Domain	Definition of Equivalence	Example
Logic	$P \equiv Q$ if $P \leftrightarrow Q$ is a tautology	$\neg(P \vee Q) \equiv \neg P \wedge \neg Q$
Set Theory	$A = B \iff \forall x \ (x \in A \leftrightarrow x \in B)$	$\{1,2,3\} = \{3,2,1\}$
Relation	'R' is an equivalence relation if reflexive, symmetric, transitive	Congruence mod 3: $a \equiv b \pmod 3$
Function	$f \equiv g$ if $f(x) = g(x)$ for all x, or structurally isomorphic	$f(x) = 2x$, $g(x) = x + x$

4.6 Contradiction

- A contradiction is a formula (or situation) that **is always false**, no matter what interpretation you give it.
- In symbols: φ is a contradiction if φ ≡ False .

  Example in logic: P ∧ ¬P.

Contradiction is "**built-in falsehood**."

In logic → unsatisfiable formulas.

In sets → impossible membership statements.

In relations → conflicting relational properties.

In functions → violations of the definition of a function.

(i) Contradiction in Logic

- A formula is contradictory if it cannot be true under any truth assignment.

Examples:

$P \land \neg P$

$(P \rightarrow Q) \land (P \land \neg Q)$

Types in Logic:

Direct contradiction – e.g. $P \land \neg P$.

Inconsistent system – e.g. axioms that lead to falsehood.

Unsatisfiable formula – e.g. $(P \land Q) \land (\neg P \lor \neg Q)$.

(ii) Contradiction in Set Theory

- A statement about sets that can never be true.

Examples:

$A \cap A^c \neq \emptyset$ (but intersection with complement is always empty).

$A \subseteq B \ \wedge \ A \nsubseteq B$ (mutually impossible).

Types in Sets:

Empty intersection → $A \cap A^c = \emptyset$.

Impossible inclusion → $A \subseteq B \wedge A \nsubseteq B$.

Cardinality contradiction → finite set equal to infinite set.

(iii) Contradiction in Relations

- A relational statement that can never hold.

Examples:

- ✓ $(a, a) \notin R \ \wedge \ (a, a) \in R$ (self-contradictory).
- ✓ Relation being both symmetric and antisymmetric in a non-trivial way:

  If $(a, b) \in R$ and $(b, a) \in R$ with $a \neq b$, symmetric requires both, antisymmetric forbids it.

Types in Relations:

Reflexive vs irreflexive contradiction → 'R' is both reflexive and irreflexive.

Symmetric vs antisymmetric contradiction → nontrivial overlap.

Impossible composition → demanding $R \circ R = \emptyset$ while 'R' is nonempty.

(iv) Contradiction in Functions

- A functional statement that is impossible by the definition of a function.

Examples:

- ✓ $f(x)=y \wedge f(x)\neq y$.
- ✓ A function mapping the same input to two different outputs.

Types in Functions:

- ✓ Well-definedness contradiction → $f(a)=b$ and $f(a)=c$ with $b\neq c$.
- ✓ Domain contradiction → demanding f(x) defined and f(x) undefined at the same time.
- ✓ Range contradiction → requiring $f(x)\in A$ and $f(x)\notin A$.

Comparison Table

Domain	Contradiction Definition	Example
Logic	Formula always false	$P \wedge \neg P$
Set Theory	Set condition impossible	$A \cap A^{c} \neq \emptyset$
Relation	Relational property impossible	'R' both symmetric and antisymmetric with $a \neq b$
Function	Mapping impossible	$f(x) = y \quad \wedge \quad f(x) \neq y$

4.7 Tautology

- A tautology is a formula (or statement) that is always true, regardless of interpretation.
- Symbolically: φ is a tautology if $\varphi \equiv$ True.
- Example in logic: $P \vee \neg P$ (Law of the Excluded Middle).

Tautology = always true.

In logic → truth laws.

In sets → complement, identity, subset properties.

In relations → reflexivity, inclusion, universal complement laws.

In functions → uniqueness and identity properties.

(i) Tautology in Logic

- A formula that evaluates to true for all truth assignments.

Examples:

$P \vee \neg P$ (law of excluded middle).

$P \rightarrow P$ (reflexivity of implication).

$(P \wedge Q) \rightarrow P$.

Types of tautologies in logic:

Logical laws → Fundamental identities (e.g., $P \vee \neg P$).

Implication tautologies → Always true conditionals (e.g., $P \rightarrow (Q \vee \neg Q)$).

Equivalence tautologies → Identity of formulas (e.g., $P \vee Q \equiv Q \vee P$).

(ii) Tautology in Set Theory

- A set statement that is always true for any set.

Examples:

$A \cup A^C = U$ (union with complement is universal set).

$A \cap U = A$.

$A \subseteq A$.

Types of tautologies in sets:

Complement tautology → $A \cup A^C = U$.

Identity tautology → $A \cap U = A$.

Subset tautology → $A \subseteq A$.

(iii) Tautology in Relations

- A relational property that is always true for any relation on a set.

Examples:

Reflexive closure ensures $(a, a) \in R$ always in closure.

$R \subseteq R$ (relation is subset of itself).

$R \cup R^C = U \times U$.

Types of tautologies in relations:

Identity tautology → $R \subseteq R$.

Complement tautology → $R \cup R^C = U \times U$.

Closure tautology → Reflexive closure always reflexive.

(iv) Tautology in Functions

- A statement about functions that is always true for any function.

Examples:

$f(x) = f(x)$.

Domain of ‘f’ is subset of domain of ‘f’.

If ‘f’ is a function, then $\forall x \in Dom(f)$, $f(x)$ is unique.

Types of tautologies in functions:

Reflexive tautology → $f(x) = f(x)$.

Domain tautology → $Dom(f) \subseteq Dom(f)$.

Well-definedness tautology → Each input has exactly one output.

Comparison Table

Domain	Tautology Definition	Example
Logic	Formula always true	$P \vee \neg P$
Set Theory	Statement always true for any set	$A \cup A^{C} = U$
Relation	Property always true for any relation	$R \cup R^{C} = U \times U$
Function	Property always true for any function	$f(x) = f(x)$

4.8 Entailment

- Entailment means one statement (or set of statements) guarantees the truth of another.
- Notation:

$$\Gamma \models \varphi.$$

Example (Logic):

From $P \wedge (P \rightarrow Q)$, we can entail Q.

Logic → entailment is truth preservation.

Sets → entailment is subset inclusion.

Relations → entailment is relation inclusion or property inheritance.

Functions → entailment is property implication (bijective → injective).

(i) Entailment in Logic

- A set of formulas 'Γ ' entails 'φ' ($\Gamma \models \varphi$) if every model that makes all formulas in 'Γ' true also makes 'φ'.

Example:

$$\{P, P \rightarrow Q\} \models Q.$$

Types of entailment in logic:

Semantic entailment ($\models$): truth-preserving across all models.

Syntactic entailment ($\vdash$): derivable using inference rules.

Material entailment: direct conditional $P \rightarrow Q$.

Pragmatic entailment: meaning-preserving in natural language

(e.g., "John killed Bill" entails "Bill is dead").

(ii) Entailment in Set Theory

- One set condition implies another set condition.

Example:

If $A \subseteq B$, then $x \in A \models x \in B$.

Symbolically:

$$x \in A \Rightarrow x \in B$$

Types in sets:

Subset entailment → $A \subseteq B$.

Equality entailment → $A=B \iff A \subseteq B \wedge B \subseteq A$

Empty set entailment → $x \in \emptyset \models$ false.

(iii) Entailment in Relations

- If one relation holds, it forces another relation to hold.

Example:

$$\text{If } R \subseteq S, \quad \text{then} \quad (a, b) \in R \models (a, b) \in S.$$

Types in relations:

Subset entailment → Relation inclusion ($R \subseteq S$).

Property entailment → Transitivity entails closure

(if 'R' is transitive, then $R \circ R \subseteq R$).

Composition entailment → If $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$.

$R \circ R$ - R of R

(iv) Entailment in Functions

- A property of a function implies another property.

Examples:

If ‘f’ is constant, it entails ‘f’ is not injective (unless domain has one element).

If ‘f’ is bijective, it entails ‘f’ is both injective and surjective.

Types in functions:

Domain entailment → $x \in Dom(f) \models f(x) \in Ran(f)$.

Property entailment → Bijective ⊨ injective and surjective.

Equality entailment → $f = g \models f(x) = g(x)$ for all ‘x’.

Dom(f) – Domain of f

bijective = injective + surjective

Comparison Table

Domain	Definition of Entailment	Example
Logic	$\Gamma \models \varphi$ if every model of 'Γ' is a model of 'φ'	$\{P, P \rightarrow Q\} \models Q$
Set Theory	Membership of one set guarantees membership of another	$A \subseteq B \implies x \in A \models x \in B$
Relations	Membership in one relation implies membership in another	$R \subseteq S \implies (a, b) \in R \models (a, b) \in S$
Functions	Functional property implies another property	Bijective $f \models$ injective and surjective

4.9 Implicature

- Implicature is an implied meaning or truth that goes beyond the literal expression.
- In logic, it's weaker than strict entailment — it depends on context or convention.
- In sets, relations, functions, we can interpret implicature as *indirect consequences* or *implicit truths* that follow from structure.

Entailment = strictly necessary truth.

Implicature = implied truth (by structure, property, or convention).

In logic → conversational and scalar implicatures.

In sets → membership implicatures.

In relations → closure/property implicatures.

In functions → property implicatures (e.g., bijection ⇒ injection + surjection).

(i) Implicature in Logic

- Inference where a statement suggests another statement without strict entailment.

Example:

"Some students passed" implicates "Not all students passed"

(but logically, "some" could mean all).

Types in logic (adapted from pragmatics):

Conventional implicature → always carried by certain connectives.

Example: "But" implicates contrast → P∧Q, with contrast implied.

Conversational implicature → inferred from cooperative principles (Grice's maxims).

Example: "John has three children" implicates "Not more than three."

Scalar implicature → choosing a weaker term implies denial of a stronger one.

"Some" implicates "Not all."

(ii) Implicature in Set Theory

- Sets can imply membership conditions beyond direct membership.

Example:

If $x \in A \cap B$, it implicates $x \in A$ and $x \in B$.

If $A \subseteq B$, it implicates "all elements of 'A' are also in 'B'."

Types in sets:

Subset implicature → $A \subseteq B$ implies membership propagation.

Union implicature → $x \in A \cup B$ implicates $x \in A$ or $x \in B$.

Intersection implicature → $x \in A \cap B$ implicates both memberships.

(iii) Implicature in Relations

- A relation can implicitly convey extra meaning.

Example:

If $(a, b) \in R$ and R is transitive, it implicates $(a, c) \in R$ if $(b, c) \in R$.

Types in relations:

Property implicature → Reflexive relation implicates $(a, a) \in R$.

Subset implicature → If $R \subseteq S$, then $(a, b) \in R$ implicates $(a, b) \in S$.

Composition implicature → If $(a, b) \in R$ and $(b, c) \in R$, it implicates $(a, c) \in R$.

(iv) Implicature in Functions

- Functions imply constraints even when not stated explicitly.

Example:

Saying “f is a function” implicates that each input has exactly one output.

If ‘f’ is bijective, it implicates that ‘f’ is also injective and surjective.

Types in functions:

Domain implicature → $x \in Dom(f)$ implicates f(x) is defined.

Codomain implicature → $f(x) \in B$ implicates mapping into codomain.

Property implicature → Bijective function implicates injective + surjective.

Comparison Table

Domain	Definition of Implicature	Example
Domain	Suggested truth not strictly entailed	"Some students passed" → implicates "Not all"
Logic	Membership/operation suggests additional truth	$x \in A \cap B$ implicates $x \in A, x \in B$
Set Theory	Structure suggests further pairs	Transitivity: $(a, b), (b, c) \in R$ implicates $(a, c) \in R$
Relations	Definition/properties imply extra truths	Bijective 'f' implicates 'f' injective & surjective

4.10 Presupposition

- A presupposition is a **background assumption that must be true for a statement to make sense** (to even have a truth value).
- A presupposition different from:

  Entailment → truth must follow logically.

  Implicature → implied but not required.

  Presupposition → required background condition.

Example (Language):

Sentence: *"The king of France is bald."*

Presupposition: *"There exists a king of France."*

Presupposition = hidden assumption that must hold to talk meaningfully.

Logic → assumes existence of individuals/definitions.

Sets → assumes sets are defined.

Relations → assumes relation structure is defined.

Functions → assumes input is in the domain and function is well-defined.

(i) Presupposition in Logic

- A condition that must hold for a formula to be meaningful or evaluable.

Example:

P(x) presupposes that ‘x’ exists in the domain.

“The father of John is kind” presupposes John has a father.

Types in logic:

Existential presupposition → variables/predicates assume existence.

P(a) presupposes ‘a’ exists.

Definitional presupposition → symbols must be defined.

Division a/b presupposes b≠0.

Structural presupposition → formula assumes a logical form is valid.

Quantifiers presuppose a nonempty domain.

(ii) Presupposition in Set Theory

- A set statement may presuppose that the sets involved are well-defined.

Examples:

$X \in A$ presupposes 'A' is a defined set.

$|A|$ presupposes 'A' is finite (if we talk about finite cardinality).

Types in sets:

Membership presupposition → talking about $x \in A$ presupposes A exists.

Operation presupposition → $A \cup B$ presupposes A,B are sets.

Cardinality presupposition → $|A|$ presupposes A is well-defined.

(iii) Presupposition in Relations

- Talking about a relation presupposes that the relation and its domain are defined.

Examples:

(a, b) ∈ R presupposes 'R' is a relation on some set.

"R is reflexive" presupposes 'R' is defined on a set.

Types in relations:

Existence presupposition → R must exist.

Domain presupposition → relation R presupposes a universe U.

Property presupposition → to check symmetry or transitivity,

'R' must already be a binary relation.

(iv) Presupposition in Functions

- A function statement presupposes the function is well-defined and its input exists in the domain.

Examples:

f(x)=y presupposes ‘f’ is defined for ‘x’.

“Inverse of ‘f’” presupposes ‘f’ is bijective.

Types in functions:

Domain presupposition → f(x) presupposes $x \in Dom(f)$.

Codomain presupposition → $f(x) \in B$ presupposes codomain ‘B’ is defined.

Inverse presupposition → f^{-1} presupposes ‘f’ is bijective.

Comparison Table

Domain	Presupposition (General Meaning)	Example
Logic	Background condition for truth evaluation	"The father of John is kind" presupposes John has a father
Set Theory	Assumption that sets are defined before use	$X \in A$ presupposes 'A' exists
Relations	Assumption that relation & domain exist	"R is reflexive" presupposes 'R' is a relation on some set
Functions	Assumption function is defined & applicable	$f(x) = y$ presupposes $x \in Dom(f)$

Thank You!